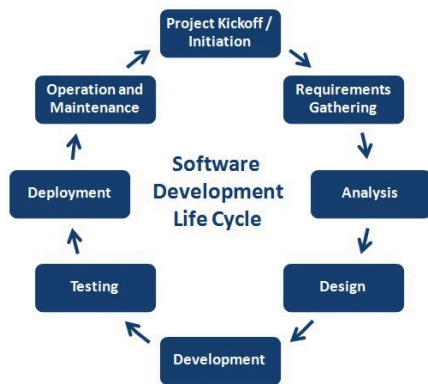


Preliminary Design Specification for NILGuard

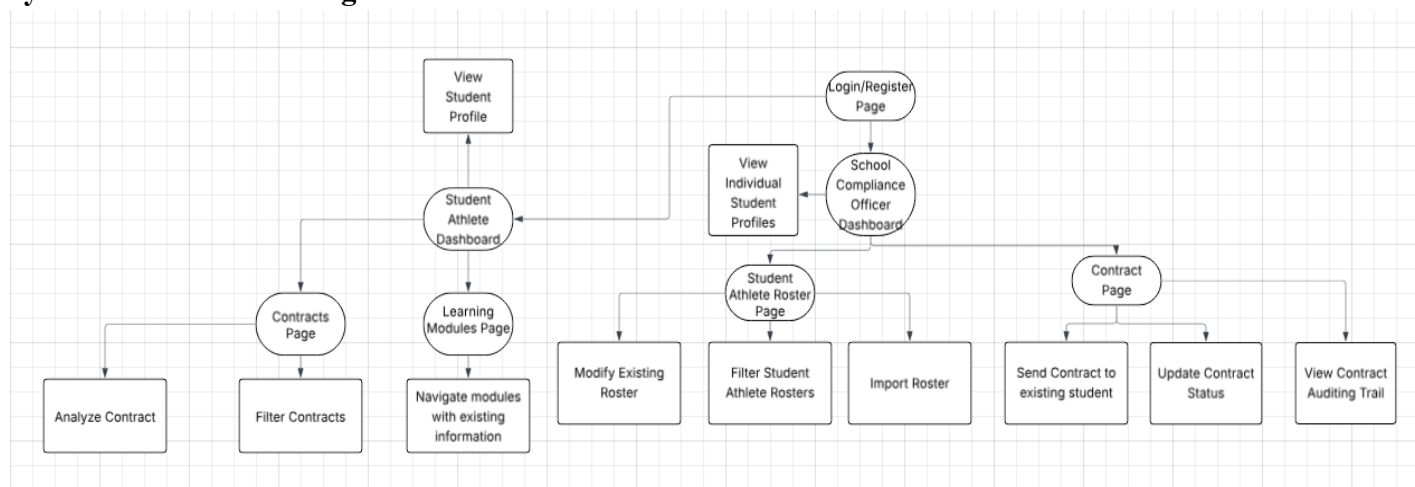


Analyze, update and submit the following documents if:

- Any new requirements not specified
- Any new change requests
- Any new updates for the schedule
- An updated budget following Gantt chart

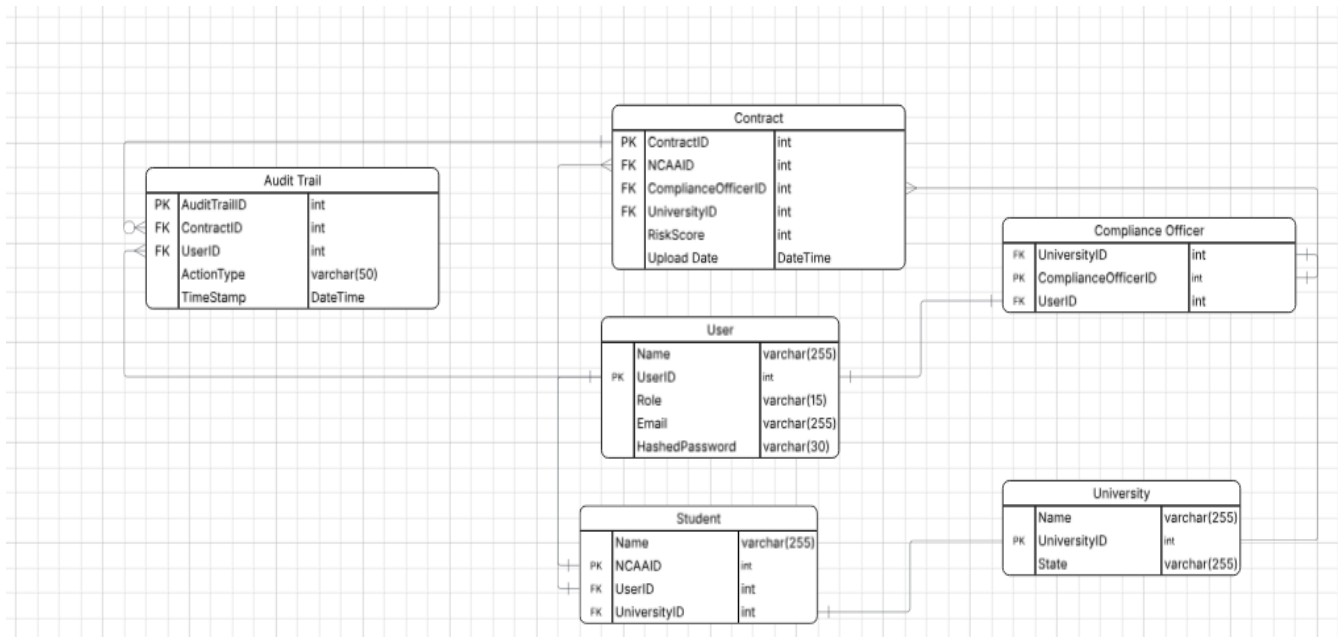
D2.	Phase2 Design and Prototype.	
D2.1.	Format Requirements: The submission file should be MS Word compatible and saved as “Project Name _part#_Your group #”, e.g. PCParts _P2_ group3. Each deliverable must contain: - Title or Project Name, - Group number, - Team lead name and - Team member names. - Sponsor Name (If Applicable) The document should be professionally formatted and free of grammar and spelling errors.	
D2.2.	Content Requirement: If OO then use Class Diagram. - Use State Transition Diagram- only if your project has dynamic behavior or workflows that are important to understand. System Design Architecture Diagram - Draw a Data Flow Diagram (DFD) of the system. You can use Visio for this task. For more information on creating DFDs, refer to the instructional videos provided in Canvas - Draw an Entity-Relationship Diagram (ERD) for the system. Any tool can be used to create this diagram.	
	Subtotal:	100 points

System Architecture Diagram



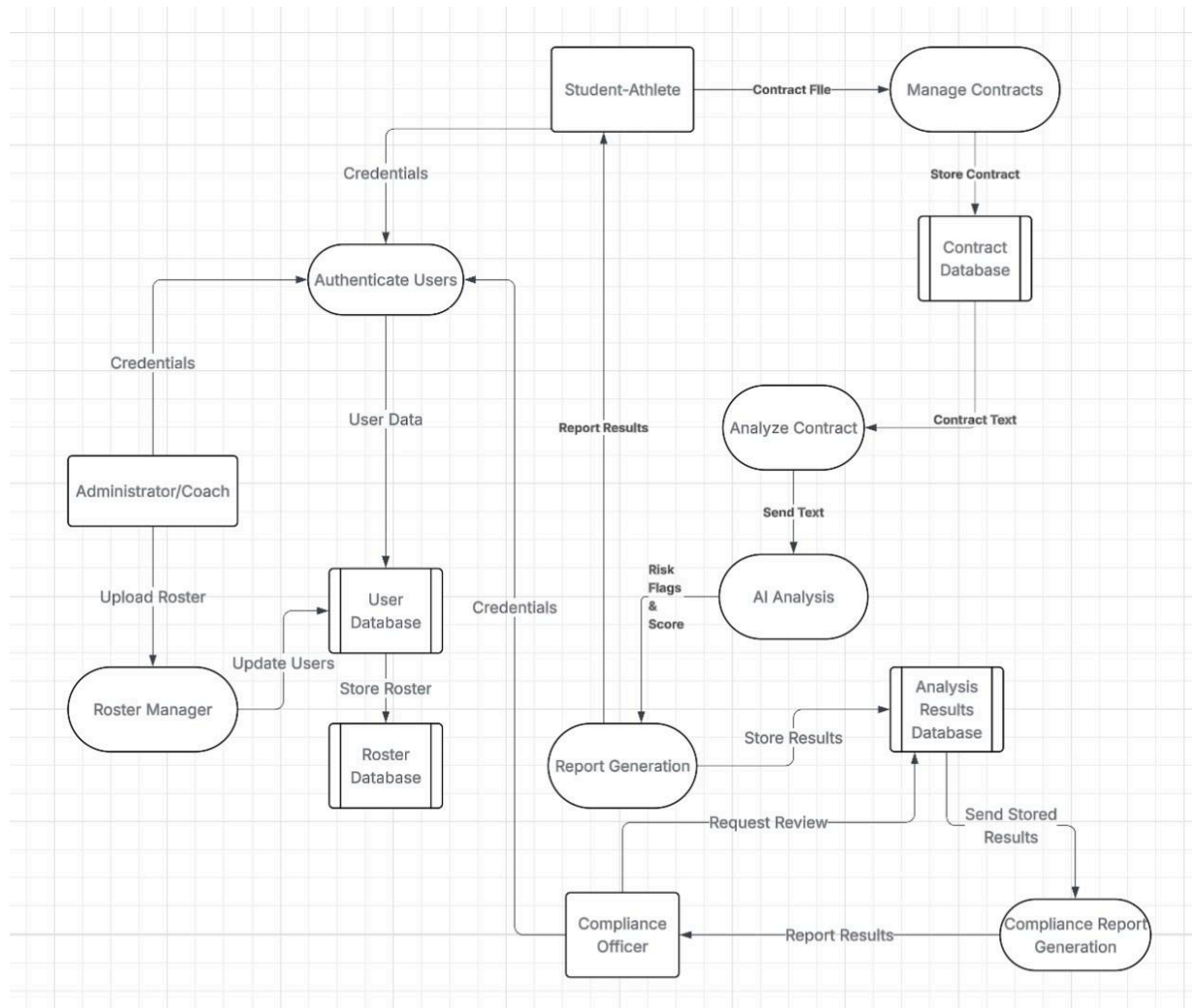
https://drive.google.com/file/d/1KkhRApu3M7glbzCyWtDKrnKdsV4S3LFb/view?usp=drive_link

Entity-Relationship Diagram (ERD)



https://drive.google.com/file/d/1KkhRApu3M7glbzCyWtDKrnKdsV4S3LFb/view?usp=drive_link

Design Flow Diagram (DFD)



Note:

- We use **DFDs** to understand data flow and processes. Great for identifying inefficiencies or missing steps in data handling.
- We use **ERDs** to design database structures. Crucial for designing a robust database to support the application's functionality.
- We use **Use Case Diagrams** to visualize system functionality from the user's perspective. Helps validate system requirements and ensures alignment with user needs.
- We use **Class Diagrams** to model the structure of object-oriented systems, offering a clear roadmap for implementation.
- Use **State Transition Diagrams** to model object behavior and state changes. Models the dynamic behavior of the system based on events and states. Captures how the system transitions between states in response to inputs. Useful for AI.